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IMAGING CONTROLS FOR UNMANNED AERIAL VEHICLES

Abstract

Methods, systems, and apparatus, including computer programs encoded on a computer storage medium, are described for performing privacy-aware surveillance with a security camera drone. The drone obtains image data using an imaging device and processes the image data to detect an object external to the drone. In response to processing the image data, the drone determines multiple viewing angles indicated in the image data with respect to the object. Based on the image data and the multiple viewing angles, a first section of the area for surveillance and a second, different section of the area to be excluded from surveillance are identified. The drone determines an adjustment to at least one of the multiple viewing angles to cause the second section to be excluded from surveillance. Based on the adjustment, the drone controls the imaging device to exclude the second section from surveillance imagery obtained by the drone.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/392,981, filed Dec. 21, 2023, which is a continuation of U.S. application Ser. No. 16/845,054, filed Apr. 9, 2020, which claims the benefit of U.S. Application Ser. No. 62/831,583, filed Apr. 9, 2019. The disclosure of each of the foregoing applications is incorporated herein by reference.

TECHNICAL FIELD

[0002] This specification generally relates to unmanned aerial devices.

BACKGROUND

[0003] Unmanned aerial vehicles can be configured as monitoring devices or sensors and are often dispersed at various locations at a property, such as a home or commercial business. These sensors and aerial devices can have distinct functions at different locations of the property. Some unmanned aerial vehicles (UAVs) at a property are autonomous vehicles or drones that offer different types of monitoring and control functionality. The functionality afforded by these drones can be leveraged to secure items at a property, to obtain information about respective items at multiple different properties, and to visually monitor conditions of items located at the properties. The usefulness and functionality of the drones may be dependent on their size, imaging ability, and related controls. In some cases, autonomous vehicles such as drones and UAVs may require small or light-weight designs that are beneficial for accomplishing objectives related to legislative initiatives, human and property safety considerations, or minimizing visual and acoustic footprints.

SUMMARY

[0004] One aspect of the subject matter described in this specification can be embodied in a method for surveilling an area using a drone. The method includes obtaining image data using an imaging device installed at the drone; processing the image data to detect an object that is external to the drone; and in response to processing the image data, determining multiple viewing angles indicated in the image data with respect to the object. The method includes, based on the image data and the multiple viewing angles, identifying a first section of the area for surveillance and a second, different section of the area to be excluded from surveillance; determining an adjustment to at least one of the multiple viewing angles to cause the second, different section of the area to be excluded from surveillance; and based on the adjustment, controlling the imaging device installed at the drone to exclude the second, different section of the area from surveillance imagery obtained by the drone.

[0005] These and other implementations can each optionally include one or more of the following features. For example, in some implementations, determining the adjustment to at least one of the multiple viewing angles includes: selecting a first viewing angle among the multiple viewing angles; calculating a new viewing angle that decreases an amount of the second, different section that is viewable from the first viewing angle; and determining the adjustment based on the new

viewing angle.

[0006] In some implementations, the method includes determining an operating altitude of the drone, wherein the operating altitude corresponds to an altitude range that allows the drone to exclude the second, different section of the area from being included in the surveillance imagery obtained by the drone; and based at least on the operating altitude, controlling the drone, a sensor installed at the drone, and the imaging device to exclude the second, different section of the area from surveillance imagery obtained by the drone. In some implementations, determining the adjustment to at least one of the multiple viewing angles includes: determining a physical attribute of the object; and determining the adjustment based on the physical attribute of the object.

[0007] The object is intermediate, or generally intermediate, the first section and the second, different section of the area. In some implementations, the object defines a boundary that separates a first property represented by the first section owned by a first entity and a second, different property represented by the second, different section owned by a second entity. In some implementations, the object is a virtual object inserted as an overlay to define a boundary between the first section and the second, different section.

[0008] Determining the adjustment to at least one of the multiple viewing angles can include determining the adjustment based on an adjustment algorithm that is implemented using the imaging device, a sensor installed at the drone, and on-board circuitry of the drone. In some implementations, the adjustment algorithm is an on-board viewing angle adjustment algorithm that is implemented using software instructions executed by a processor integrated in the on-board circuitry of the drone.

[0009] The adjustment algorithm is operable to adjust viewing angles of the imaging device during operation of the drone as the drone obtains surveillance imagery of the property. In some implementations, the method includes performing an initial mapping of the area based on a mapping algorithm; generating an initial map of the area in response to performing the initial mapping; and generating, using the mapping algorithm, one or more subsequent maps of the area based on the adjustment.

[0010] The drone can include multiple cameras, and each camera of the multiple cameras can represent a respective imaging device installed at the drone. In some implementations, processing the image data includes determining frame coverages for each of the multiple cameras installed at the drone. Determining the frame coverages can include dynamically calculating a respective frame coverage for each of multiple cameras installed at the drone, wherein each of the respective frame coverages are calculated concurrently as the image data is being obtained.

[0011] Other implementations of this and other aspects include corresponding systems, apparatus, and computer programs, configured to perform the actions of the methods, encoded on computer storage devices. A system of one or more computers can be so configured by virtue of software, firmware, hardware, or a combination of them installed on the system that in operation cause the system to perform the actions. One or more computer programs can be so configured by virtue of having instructions that, when executed by data processing apparatus, cause the apparatus to perform the actions.

[0012] The details of one or more implementations of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1. shows an example of a system that includes an example drone used to implement

privacy-aware surveillance.

[0014] FIG. 2 shows an example drone adjusting one or more viewing angles of an imaging device installed at the drone.

[0015] FIG. 3 shows an example of a drone performing surveillance of an object at an area or property.

[0016] FIG. 4 shows an example of a frame that is captured using an imaging device installed at a drone.

[0017] FIG. 5 shows a block diagram of an example property and property monitoring system that includes drones for monitoring conditions at the property.

[0018] FIG. 6 shows an example process for monitoring a property using a drone.

[0019] FIG. 7 shows a diagram illustrating an example property monitoring system.

[0020] Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0021] This specification describes techniques for improved control of the imaging capabilities of unmanned, autonomous aerial vehicles (“drones”) as well as property monitoring systems for commercial or residential properties that can use the drones to monitor items or conditions at the property. For example, a drone is described that includes at least an imaging device and one or more sensors. The drone can represent an example security drone or security camera drone system. In general, an example security camera drone system may be installed and configured to operate and perform surveillance in many different environments or areas, including areas with multiple different properties.

[0022] A drone may be tasked to monitor or perform surveillance on a property in area where the property is located close to, or substantially to, another property (described below as a “third-party property”). The drone includes a digital imaging device, such as a digital camera, and on-board circuitry that are used to capture images during surveillance of the property. Based on the described techniques, the imaging device and on-board circuitry of the drone can be configured to dynamically adjust an altitude of the drone and viewing angles employed by the imaging device to exclude, avoid, or otherwise reduce a likelihood of capturing images and video of the other property in surveillance imagery obtained by the drone.

[0023] The property monitoring system may include sensors, such as motion or imaging sensors, at a property to monitor conditions at the property. In addition to the sensors, drones can be positioned at the property to monitor conditions at the property. The monitoring system includes a control unit and controls that enable automation of actions at the property. Security/property monitoring systems can include several sensors, drones, and other devices that perform specific functions at a property. These drones and devices interact with the control units, or directly with a monitoring server, to provide sensor data to the monitoring server and to receive commands from the monitoring server.

[0024] FIG. 1. shows an example of a system **100** that includes a drone **110** used to implement privacy-aware surveillance for an example surveilled area. The drone **110** can represent an example security drone and the system **100** can represent an example security camera drone system.

[0025] In the present disclosure, an example area to be surveilled by a security camera drone may be referred to as a “surveilled area.” The surveilled area **120** may be located next to, adjacent to, or within a threshold proximity of another area that includes (or is represented by) an example “third-party property.” In some examples, the surveilled area **120** is referred to as a first area **120** and the other area **130** that includes the third-party property **135** is referred to as a second, different area **130**, relative to the first area **120**. In some other examples, an area (e.g., the surveilled area) includes multiple properties and a first property **125** to be surveilled or monitored using the drone may be referred to as a first section **120** of the area, whereas a second property **135** (e.g., third-party property) that is not being surveilled or monitored may be referred to as a second section of the area that is different than the first section.

[0026] A first property, section, or area to be surveilled may be owned by an entity or individual that is different than an entity or individual that owns a second, different property, section, or area that is not being surveilled. For example, a first property **125** being surveilled may have a different owner than an adjacent second property **135** that is not being surveilled. In some implementations, the property **125** is an item under surveillance, is represented by the first section **120**, and is owned by a particular entity. Similarly, the property **135** can be represented by the section **130**, and is owned by a different entity.

[0027] At least because of this difference in ownership, the owner of the property **125** being surveilled, including the owner or operator of the drone **110**, may not have an interest in surveilling the other property **135**. In some instances, the owner of the property **125** being surveilled may want to avoid viewing the other property **135** and/or capturing images or videos of the other property **135**. In some cases, an owner of the other property **135** may have a privacy interest in that property, such that the owner of the other property **135** does not want a drone **110** capturing or recording images and video of the property.

[0028] The system **100** is configured to obtain or determine, as well as store, information about the surveilled area **120** and the third-party property **135**. For example, the information can correspond to imagery data and video data that are obtained and then stored at the drone **110** for further processing. Although not shown in the example of FIG. 1, in addition to the drone **110**, the system **100** may correspond to an example property monitoring system that includes one or more control units and a monitoring server. In some implementations, the data obtained using the drone **110** may be processed locally at the drone, remotely at the monitoring server, or both. This is described in more detail below with reference to the example of FIG. 5.

[0029] The system **100** is configured to determine a physical attribute of objects in an area. This area may be a large area that includes the smaller surveilled area/section **120** that has the property **125** and the other smaller area/section **130** that has the third-party property **135**. An example object **140** may be located intermediate each of these first and second areas. The object **140** may define a boundary that separates property **125** and property **135**. For example, the drone **110** may be configured to determine a height of an object **140** such as fence that is located between the surveilled area **120** and the other area **130** or the third-party property **135**. The drone **110** can determine the attribute, height, or size of the fence or object at various points along the fence. In some implementations, the object **140** is a virtual object that is inserted as an overlay to define a boundary between the first section **120** and the second, different section **130**.

[0030] The drone **110** can detect, recognize, and estimate the attributes (e.g., height) of the object **140** during an initial mapping of the surveilled area **120** or during a surveillance mission. For example, the drone **110** determines the attribute by using on-board imaging devices, such as cameras, and sensors. In some implementations, the drone **110** determines the attribute based on one or more algorithms that may be executed or implemented locally at the drone **110**. Additionally or alternatively, an installer or operator of the drone **110** may provide the attribute of the object, such as a height along a perimeter of the area, as a manual input to drone **110**. For example, the operator may provide the height input using an application program that is operable to push user inputs to the drone's on-board processors for local processing by the drone **110**.

[0031] The drone **110** includes on-board circuitry that is coupled to the sensing/imaging devices installed at the drone, including hardware circuits and special-purpose processors such as processors that run machine-learning models that are implemented using trained neural networks. The on-board circuitry generates control signals that are routed to sensing, imaging, and/or data processing devices of the drone **110** to cause these devices to perform one or more functions. For example, the control signals can be used to control various functions of the drone **110**, such as controlling a flight path of the drone or causing the imaging device **115** to obtain image data. In some implementations, the imaging device **115** obtains the image data based on control signals that are generated in accordance with an example mapping algorithm, an adjustment algorithm, or both.

[0032] Information obtained by system **100** can include sets of parameters relating to operations of various sensors **115** and/or devices **115** installed at the drone **110**. For example, the sensors are operable to transmit signals (e.g., radio signals, optical signals, etc.) that can be used to control a flight path of the drone **110**. To control its autonomous operations drone **110** can include a number of sensors or sensing devices **115**. For example, the sensing devices **115** can include mono, stereo or time-of-flight (ToF) cameras, rangefinders, and other related devices. In some implementations, a sensor **115** installed at the drone **110** is a transceiver device is operable to transmit a radio (or optical) signal and to detect radio (or optical), including signals that are based on an earlier transmitted signal.

[0033] The parameters obtained by the system **100** may include information about the field-of-view (“FOV”) of one or more cameras or imaging devices/sensors **115** of the drone **110**. In some implementations, the system **100** determines the viewing angles based on an observed field-of-view of the camera **115**. For example, the system **100** determines a respective field-of-view (FOV) of at least one camera **115** installed at the drone **110** and then determines a respective set of multiple viewing angles for each FOV observed for each camera.

[0034] In some implementations, the drone **110** parameters include an operating altitude range of the drone **110**. In some other implementations, for one or more sets of parameters obtained for the on-board cameras and sensors **115**, the system **100** is operable to obtain, determine, or otherwise derive information about sets of images or streams of video data that were captured for multiple different camera angles and with reference to different fields-of-views of the cameras.

[0035] In the example of FIG. 1 a drone **110** may hover above the surveilled area **120**. As the drone hovers, an example default unadjusted pitch angle (e.g., a viewing angle) of the camera **115** is such that an example FOV **150** of an drone **110** camera allows the capture of the surveilled area **120** and a portion of the property **130**. To decrease the amount of the property **130** in the FOV **150**, the drone **110** (or system **100**) implements one or more software algorithms to calculate a new pitch angle of the camera **115** and adjusts the camera **115** to the new pitch angle so that at least an upper boundary of an adjusted FOV **160** is aligned or positioned below a height of the object **140**.

[0036] FIG. 2 shows an example drone **110** adjusting one or more viewing angles of an imaging device **115** relative to different flight altitudes. To perform the surveillance, the drone **110** may fly above a particular level of an object **140** in the area. As described above, the object **140** may be an item such as a fence or physical boundary that separates the two properties **125**, **135**. The drone **110** may employ certain predefined or preset viewing angles or be positioned at a certain altitude that results in the drone capturing images and/or video of the adjacent property **135**, including persons located at the other property.

[0037] In view of this, the techniques described in this document can be used to alleviate or mitigate privacy concerns related to improper or unauthorized surveillance of individuals and properties. For example, the imaging device and on-board circuitry of the drone are configured to dynamically adjust the altitude and viewing angles used by the drone to exclude images and video of the adjacent property from surveillance imagery obtained for the property being surveilled. Accordingly, the present disclosure describes methods that can be used to increase third-party privacy, for example, by adjusting the imaging angles of the drone or imaging device, as well as editing images or videos captured by the drone.

[0038] In some implementations, the drone **110** includes a multiple cameras, and each camera of the multiple cameras can represent a respective imaging device installed at the drone **110**. The drone **110** is operable to obtain image data using the cameras and process the image data. In some implementations, processing the image data includes determining frame coverages for each of the multiple cameras installed at the drone **110**. For example, when flying over the surveilled area **120**, the drone **110** may dynamically calculate frame coverage of its cameras and/or sensors **115** and adjust the angle of the cameras and/or sensors **115** and the altitude of the drone **110** accordingly to decrease the amount of the property **130** that is viewable in a FOV.

[0039] Determining the frame coverages includes the drone **110** dynamically calculating a respective frame coverage for each of the multiple cameras installed at the drone **110**. In some implementations, wherein each of the respective frame coverages are calculated concurrently as the image data is being obtained. The frame coverage can indicate an amount or type of items in the area that are included in a captured image or video frame. In some implementations, the system **100** determines an operating altitude of the drone **110**, where the operating altitude corresponds to an altitude range that allows the drone to exclude the second, different section **130** of the area from being included in the surveillance imagery obtained by the drone **110**. Based at least on the operating altitude, the system **100** is operable to control the drone **110** and sensors/imaging device **115** installed at the drone to exclude the section **130** from the surveillance imagery.

[0040] Additionally or alternatively, the drone **110** (or system **100**) may be configured to adjust the position, orientation, and/or altitude of the drone **110** to reduce the amount of the property **135** in the FOV of the cameras **115** of the drone **110**. In the example of FIG. 2, as the drone **110** is hovering or flying above the surveilled area **120**, the drone **110** may use algorithms to calculate a new position **170** and reposition the drone **110** accordingly so that the upper boundary of the adjusted FOV **160** is aligned or positioned below the height of the object **140**.

[0041] The system **100**, including drone **110**, calculates the new position **170** based on an adjustment algorithm that is implemented using the on-board circuitry of the drone as well as the imaging devices and sensors installed at the drone **110**. In some implementations, the adjustment algorithm is an on-board viewing angle adjustment algorithm that is implemented using software instructions executed by a processor integrated in the on-board circuitry of the drone. The adjustment algorithm is described in more detail below with reference to FIG. 4.

[0042] FIG. 3 shows an example of a drone performing surveillance of an object **140** at an area or property. In some cases, the drone **110** may be configured to capture images and/or video of tall objects **145**, for example, buildings that are located on the surveilled area **120**. In such cases, it may be difficult to exclude property **135** from the FOV **150**, and/or the frame of drone **110** cameras and/or sensors **115**. In such cases, algorithms implemented at the drone **110** may be used to calculate and apply a digital mask over captured frames or images to render property **135** invisible or obscured. Additionally or alternatively, the drone **110** software algorithms may be configured to erase recorded images and/or video of the property **135**.

[0043] FIG. 4 shows an example of a frame **400** that is captured using an imaging device installed at a drone. The software/adjustment algorithms implemented at the drone **110** can be used to determine which pixels in each captured frame boundary **410** correspond to one or more of: the surveilled area **120**, surveilled objects/property **130**, and/or the object **140** along the perimeter of the surveilled area.

[0044] The system **100** can process the pixilation data and pixel images values against the map of the area and based on the processing, determine that other pixels **450** in a given frame do not belong to the surveilled property, but rather potentially contain information about the property **130**. Additionally or alternatively, the drone **110** may determine, based on an adjustment algorithm, which pixels of a given frame are part of the property **130** directly. In this manner, pixels potentially containing information about the property **130** may then be obscured, masked, or erased to decrease the amount of information about the third-party property **130** that is stored by the drone **110** or system **100**.

[0045] The drone **110** is configured to compute, using the adjustment algorithms, determinations about the pixels based on the information about surveilled area perimeter as well as depth data (e.g., in the form of point-cloud) about the objects located inside the surveilled area **120**, **130**. The information about the surveilled areas and the depth data may be collected, for example, during initial mapping of the surveilled area as described above. In some implementations, information about the surveilled area may be provided by a user or installer of the drone **110**. For example, an operator or installer may view images recorded by the drone **110** and select portions that are of the

surveilled area and portions that are of the property **130**.

[0046] In some other implementations, drone **110** is configured to use one or more machine-learning models to detect, classify, and/or identify objects in a frame and/or parts of the frame that belong to the surveilled area **120** as well as parts of the frame **400** that do not belong to the surveilled area **120**. This information may be used to apply a mask to certain parts of the frame as described above with reference to FIG. 3.

[0047] FIG. 5 shows a block diagram of an example property and property monitoring system (“system **500**”) that includes drones for monitoring conditions at the property. The property **502** may be, for example, a residence, such as a single family home, a townhouse, a condominium, or an apartment. In some examples, the property **502** may be a commercial property, a place of business, or a public property, such as a police station, fire department, or military installation. The property **502** may correspond to the property **125** described above.

[0048] The system **500** can include multiple sensors **520**. Each sensor **520** can be associated with various types of devices that are located at property **502**. For example, a sensor(s) can be associated with contact sensor **520a** that is operable to detect when a door or window is opened or closed. In some examples, a sensor can be a temperature control device **520b** that is operable to control a temperature setting for an area or room at the property **502**. Similarly, a sensor can be associated with a video or image recording device **520c** located at the property **502**, such as a digital camera or other electronic recording device.

[0049] As described above, the property **502**, **125** is monitored by a property monitoring system. The property monitoring system includes a control unit **510** that sends sensor data **525**, obtained using sensors **520**, to a remote monitoring server **560**. In some implementations, the control units, monitoring servers, or other computing modules described herein are included as sub-systems of the property monitoring system **500**. Each of the sensors **520** can use various types of technology to transmit sensor signal data or to exchange data communications with devices of system **500** (or the property monitoring system). In some implementations, one or more sensors **520** at the property **502** can be at least one of: a Z-Wave enabled sensing device, a Bluetooth enabled sensing device, a Wi-Fi enabled sensing device, or a sensing device that uses radio or wireless signal technology. Additional sensor features are described in more detail below.

[0050] The system **500** also includes a drone **110** that is operable to generate image data which may be transmitted as sensor data **525** to the monitoring server **560**. The image data can be processed at the monitoring server **560** to monitor and determine conditions at the property or to generate a layout or mapping of items or objects, including sensors **520**, at the property **502**. For example, a drone **110** can be used to survey the property **502** to determine whether items are secure at the property or to build a multi-dimensional map, such as a two-dimensional (2D) or three-dimensional (3D) model that includes 2D/3D map or physical layout of the property **502**. In some implementations, the drone **110** surveys the property **502** and transmits parameters (or parameter values), sensor/drone data, and status information to the monitoring server **560** to determine conditions at the property or to generate the multi-dimensional map/model.

[0051] For example, the drone **110** is operable to transmit: (i) parameter signals that characterize a status or attribute of the items/objects, (ii) data that describes communications and interactions between the drone **110** and each of the sensors **520** or items at the property **502**, and (iii) imagery data (e.g., “mapping data”) about the property **502** to the monitoring server **560** to generate the multi-dimensional model. In some examples, the drone **110** interacts with the monitoring server **560** to determine the conditions concurrent with the drone **110** performing the survey of property **502**.

[0052] The drone **110** are operable to receive commands **570** and other signals from the monitoring server **560** or from a client device **540** of a user **508**. In some implementations, the command **570** is a data payload that includes instructions and data values for commanding the drone **110** to perform one or more specific functions at the property **502**. For example, a command **570** can cause the

drone **110** to travel to a particular area of the property **502**, triggering of a particular sensor **520** at the property **502**, or transmitting imagery or mapping data from the drone **110** to the monitoring server **560**.

[0053] FIG. **6** shows an example process **600** for monitoring a property using a drone. The drone can correspond to an example drone described above, including a security drone or a security camera drone. Process **600** can be implemented or executed using the system **100**, **400** described above. Descriptions of process **600** may reference the above-mentioned computing resources of system **100**, **500**. The steps or actions of process **600** may be enabled by programmed firmware, or software instructions, that are executable by one or more processors of the devices and resources described in this document. The process **600** can correspond to method for surveilling an area that includes two or more properties using a drone. In some implementations, the drone is a security camera drone and the steps of process **600** correspond to a method for performing privacy-aware surveillance with a security camera drone.

[0054] Referring now to process **600**, the process includes obtaining image data using an imaging device installed at a drone (**610**). For example, the image data can be obtained by the camera/imaging device **115** for processing by a processor installed at on-board circuitry of the drone **110**. In some implementations, the system **500** provides commands or control signals to the drone **110** to cause the camera/imaging device **115** to obtain (or capture) the image data. In some implementations, the process **600** includes one or more sensors transmitting signals that are used to control a flight path of the drone or operation associated with surveilling or monitoring a property in an area.

[0055] For example, an exterior sidewall of the drone **110** is configured to receive or support one or more sensors **115**. The sensors may be represented by an imaging device **115** or an example transceiver **115** that includes a transmitter and a receiver for transmitting and detecting/receiving signals, respectively. In some implementations, the sensors **115** of the drone **110** are operable to transmit and receive signals based on commands or control signals received from the on-board circuitry of the drone, from the monitoring server **560** described above, or both.

[0056] In some cases, the monitoring server **560** of system **500** provides commands to the drone **110** to cause the sensors **115** to transmit and receive signals for controlling imaging functions of the drone **110**, controlling a flight path of the drone **110**, or both. The signals can be radio signals, optical signals, or various other types of signals that can be used to control the flight path of drone **110** as well as to control viewing angles of cameras or imaging devices installed at the drone. The sensors **115** may be designed or configured to include the components and circuitry required to transmit and detect one or more of the different types of signals that can be used to control the drone **110**.

[0057] Process **600** includes processing at least the image data to detect one or more objects that are external to the drone (**620**). For example, the image data, including other sensor data collected by the drone **110**, is processed using the on-board circuitry of the drone **110**, the monitoring server **560**, or both. For example, the data may be processed locally at the drone **110** to control a specific operation performed by the drone, or remotely at an example control engine of the monitoring server **560**. The image data and other sensor data are processed to detect objects that are external to the drone during aerial maneuvering of the drone **330** or when the drone travels around a property to monitor conditions items and objects at the property.

[0058] In response to processing the image data, the drone **110** identifies or determines different viewing angles that are indicated in the image data with respect to the object (**630**). For example, the drone **110** can identify or determine the viewing angles based on parameters or settings of the imaging lens of the cameras used to capture the images. In some implementations, the drone determines the viewing angles at least by using image recognition to identify the object and other items in an image, processing information about the objects/items (e.g., positional/coordinate data for the objects) against a map of the area, and then determine the viewing angles based on the

processing.

[0059] Based on the image data and the multiple viewing angles, the drone **110** identifies a first section of the area for surveillance and a second, different section of the area to be excluded from surveillance (**640**). For example, the drone **110** can identify the first section **120** that includes property **125** as a section of the area that requires surveillances and can identify the second section **130** that includes property **135** as a section of the area that is to be excluded from surveillance. In some implementations, the drone **110** can identify the different sections based on at least one user inputs that are processed against the images of the viewing angles, the mapping of the area, or both.

[0060] The drone **110** determines an adjustment to at least one of the multiple viewing angles to cause the second, different section **130** of the area to be excluded from surveillance imagery obtained by the drone (**650**). For example, the drone **110** can determine the adjustment to one or more one of the multiple viewing angles at least by selecting a first viewing angle among the multiple viewing angles, calculating a new viewing angle that decreases an amount of the section **130** that is viewable from the first viewing angle, and determining the adjustment based on the new viewing angle.

[0061] In some implementations, the drone **110** may perform a similar calculation for each of multiple viewing angles. The new angle is used to adjust an angle of the camera **115** to a new pitch angle so that at least an upper boundary of an adjusted FOV **160** is aligned or positioned below a height of the object **140**. Based on the adjustment, the drone **110** controls the imaging device installed at the drone to exclude the second, different section **130** of the area from surveillance imagery obtained by the drone (**660**).

[0062] FIG. 7 is a diagram illustrating an example of a property monitoring system **700**. The electronic system **700** includes a network **705**, a control unit **710**, one or more user devices **740** and **750**, a monitoring server **760**, and a central alarm station server **770**. In some examples, the network **705** facilitates communications between the control unit **710**, the one or more user devices **740** and **750**, the monitoring server **760**, and the central alarm station server **770**.

[0063] The network **705** is configured to enable exchange of electronic communications between devices connected to the network **705**. For example, the network **705** may be configured to enable exchange of electronic communications between the control unit **710**, the one or more user devices **740** and **750**, the monitoring server **760**, and the central alarm station server **770**. The network **705** may include, for example, one or more of the Internet, Wide Area Networks (WANs), Local Area Networks (LANs), analog or digital wired and wireless telephone networks (e.g., a public switched telephone network (PSTN), Integrated Services Digital Network (ISDN), a cellular network, and Digital Subscriber Line (DSL)), radio, television, cable, satellite, or any other delivery or tunneling mechanism for carrying data. Network **705** may include multiple networks or subnetworks, each of which may include, for example, a wired or wireless data pathway. The network **705** may include a circuit-switched network, a packet-switched data network, or any other network able to carry electronic communications (e.g., data or voice communications). For example, the network **705** may include networks based on the Internet protocol (IP), asynchronous transfer mode (ATM), the PSTN, packet-switched networks based on IP, x.25, or Frame Relay, or other comparable technologies and may support voice using, for example, VoIP, or other comparable protocols used for voice communications. The network **705** may include one or more networks that include wireless data channels and wireless voice channels. The network **705** may be a wireless network, a broadband network, or a combination of networks including a wireless network and a broadband network.

[0064] The control unit **710** includes a controller **712** and a network module **714**. The controller **712** is configured to control a control unit monitoring system (e.g., a control unit system) that includes the control unit **710**. In some examples, the controller **712** may include a processor or other control circuitry configured to execute instructions of a program that controls operation of a control unit system. In these examples, the controller **712** may be configured to receive input from

sensors, flow meters, or other devices included in the control unit system and control operations of devices included in the household (e.g., speakers, lights, doors, etc.). For example, the controller **712** may be configured to control operation of the network module **714** included in the control unit **710**.

[0065] The network module **714** is a communication device configured to exchange communications over the network **705**. The network module **714** may be a wireless communication module configured to exchange wireless communications over the network **705**. For example, the network module **714** may be a wireless communication device configured to exchange communications over a wireless data channel and a wireless voice channel. In this example, the network module **714** may transmit alarm data over a wireless data channel and establish a two-way voice communication session over a wireless voice channel. The wireless communication device may include one or more of a LTE module, a GSM module, a radio modem, cellular transmission module, or any type of module configured to exchange communications in one of the following formats: LTE, GSM or GPRS, CDMA, EDGE or EGPRS, EV-DO or EVDO, UMTS, or IP.

[0066] The network module **714** also may be a wired communication module configured to exchange communications over the network **705** using a wired connection. For instance, the network module **714** may be a modem, a network interface card, or another type of network interface device. The network module **714** may be an Ethernet network card configured to enable the control unit **710** to communicate over a local area network and/or the Internet. The network module **714** also may be a voice band modem configured to enable the alarm panel to communicate over the telephone lines of Plain Old Telephone Systems (POTS).

[0067] The control unit system that includes the control unit **710** includes one or more sensors. For example, the monitoring system may include multiple sensors **720**. The sensors **720** may include a lock sensor, a contact sensor, a motion sensor, or any other type of sensor included in a control unit system. The sensors **720** also may include an environmental sensor, such as a temperature sensor, a water sensor, a rain sensor, a wind sensor, a light sensor, a smoke detector, a carbon monoxide detector, an air quality sensor, etc. The sensors **720** further may include a health monitoring sensor, such as a prescription bottle sensor that monitors taking of prescriptions, a blood pressure sensor, a blood sugar sensor, a bed mat configured to sense presence of liquid (e.g., bodily fluids) on the bed mat, etc. In some examples, the health monitoring sensor can be a wearable sensor that attaches to a user in the home. The health monitoring sensor can collect various health data, including pulse, heart-rate, respiration rate, sugar or glucose level, bodily temperature, or motion data.

[0068] The sensors **720** can also include a radio-frequency identification (RFID) sensor that identifies a particular article that includes a pre-assigned RFID tag.

[0069] The control unit **710** communicates with the home automation controls **722** and a camera **730** to perform monitoring. The home automation controls **722** are connected to one or more devices that enable automation of actions in the home. For instance, the home automation controls **722** may be connected to one or more lighting systems and may be configured to control operation of the one or more lighting systems. Also, the home automation controls **722** may be connected to one or more electronic locks at the home and may be configured to control operation of the one or more electronic locks (e.g., control Z-Wave locks using wireless communications in the Z-Wave protocol). Further, the home automation controls **722** may be connected to one or more appliances at the home and may be configured to control operation of the one or more appliances. The home automation controls **722** may include multiple modules that are each specific to the type of device being controlled in an automated manner. The home automation controls **722** may control the one or more devices based on commands received from the control unit **710**. For instance, the home automation controls **722** may cause a lighting system to illuminate an area to provide a better image of the area when captured by a camera **730**.

[0070] The camera **730** may be a video/photographic camera or other type of optical sensing device configured to capture images. For instance, the camera **730** may be configured to capture images of

an area within a building or home monitored by the control unit **710**. The camera **730** may be configured to capture single, static images of the area and also video images of the area in which multiple images of the area are captured at a relatively high frequency (e.g., thirty images per second). The camera **730** may be controlled based on commands received from the control unit **710**.

[0071] The camera **730** may be triggered by several different types of techniques. For instance, a Passive Infra-Red (PIR) motion sensor may be built into the camera **730** and used to trigger the camera **730** to capture one or more images when motion is detected. The camera **730** also may include a microwave motion sensor built into the camera and used to trigger the camera **730** to capture one or more images when motion is detected. The camera **730** may have a “normally open” or “normally closed” digital input that can trigger capture of one or more images when external sensors (e.g., the sensors **720**, PIR, door/window, etc.) detect motion or other events. In some implementations, the camera **730** receives a command to capture an image when external devices detect motion or another potential alarm event. The camera **730** may receive the command from the controller **712** or directly from one of the sensors **720**.

[0072] In some examples, the camera **730** triggers integrated or external illuminators (e.g., Infra-Red, Z-wave controlled “white” lights, lights controlled by the home automation controls **722**, etc.) to improve image quality when the scene is dark. An integrated or separate light sensor may be used to determine if illumination is desired and may result in increased image quality.

[0073] The camera **730** may be programmed with any combination of time/day schedules, system “arming state”, or other variables to determine whether images should be captured or not when triggers occur. The camera **730** may enter a low-power mode when not capturing images. In this case, the camera **730** may wake periodically to check for inbound messages from the controller **712**. The camera **730** may be powered by internal, replaceable batteries if located remotely from the control unit **710**. The camera **730** may employ a small solar cell to recharge the battery when light is available. Alternatively, the camera **730** may be powered by the controller's **712** power supply if the camera **730** is co-located with the controller **712**.

[0074] In some implementations, the camera **730** communicates directly with the monitoring server **760** over the Internet. In these implementations, image data captured by the camera **730** does not pass through the control unit **710** and the camera **730** receives commands related to operation from the monitoring server **760**.

[0075] The system **700** also includes thermostat **734** to perform dynamic environmental control at the home. The thermostat **734** is configured to monitor temperature and/or energy consumption of an HVAC system associated with the thermostat **734**, and is further configured to provide control of environmental (e.g., temperature) settings. In some implementations, the thermostat **734** can additionally or alternatively receive data relating to activity at a home and/or environmental data at a home, e.g., at various locations indoors and outdoors at the home. The thermostat **734** can directly measure energy consumption of the HVAC system associated with the thermostat, or can estimate energy consumption of the HVAC system associated with the thermostat **734**, for example, based on detected usage of one or more components of the HVAC system associated with the thermostat **734**. The thermostat **734** can communicate temperature and/or energy monitoring information to or from the control unit **710** and can control the environmental (e.g., temperature) settings based on commands received from the control unit **710**.

[0076] In some implementations, the thermostat **734** is a dynamically programmable thermostat and can be integrated with the control unit **710**. For example, the dynamically programmable thermostat **734** can include the control unit **710**, e.g., as an internal component to the dynamically programmable thermostat **734**. In addition, the control unit **710** can be a gateway device that communicates with the dynamically programmable thermostat **734**. In some implementations, the thermostat **734** is controlled via one or more home automation controls **722**.

[0077] A module **737** is connected to one or more components of an HVAC system associated with

a home, and is configured to control operation of the one or more components of the HVAC system. In some implementations, the module **737** is also configured to monitor energy consumption of the HVAC system components, for example, by directly measuring the energy consumption of the HVAC system components or by estimating the energy usage of the one or more HVAC system components based on detecting usage of components of the HVAC system. The module **737** can communicate energy monitoring information **576** and the state of the HVAC system components to the thermostat **734** and can control the one or more components of the HVAC system based on commands received from the thermostat **734**.

[0078] The system **700** includes one or more UAV control engines **757**. Each of the one or more UAV control engine **757** connects to control unit **710**, e.g., through network **705**. The UAV control engines **757** can be computing devices (e.g., a computer, microcontroller, FPGA, ASIC, or other device capable of electronic computation) capable of receiving data related to the sensors **720** and communicating electronically with the monitoring system control unit **710** and monitoring server **760**. In some implementations, the sensors **720** correspond to the sensors or an imaging device (e.g., a camera) that is installed in an example drone, such as in the internal cavity of a frame of the drone.

[0079] The UAV control engine **757** receives data from one or more sensors **720**. In some examples, the UAV control engine **757** can be used to control a drone at the property **502**. The UAV control engine **757** can be used to control a flight path of the drone based on data generated by sensors **720**, e.g., data from sensor **720** describing motion, movement, acceleration/velocity, orientation, and other parameters, pertaining to a drone used to monitor conditions at the property **502**. The UAV control engine **757** can receive data from the one or more sensors **720** through any combination of wired and/or wireless data links. For example, the UAV control engine **757** can receive sensor data via a Bluetooth, Bluetooth LE, Z-wave, or Zigbee data link.

[0080] The UAV control engine **757** communicates electronically with the control unit **710**. For example, the UAV control engine **757** can send data related to the sensors **720** to the control unit **710** and receive commands related to determining a state of a drone at the property **502** and a flight path of the drone based on data from the sensors **720**. In some examples, the UAV control engine **757** processes or generates sensor signal data, for signals emitted by the sensors **720**, prior to sending it to the control unit **710**. The sensor signal data can include information that indicates a user **508** has interacted with a drone at the property or is using a drone to monitor items at the property.

[0081] In some examples, the system **700** further includes one or more robotic devices **790**. The robotic devices **790** may be any type of robots that are capable of moving and taking actions that assist in home monitoring. For example, the robotic devices **790** may include drones that are capable of moving throughout a home based on automated control technology and/or user input control provided by a user. In this example, the drones may be able to fly, roll, walk, or otherwise move about the home. The drones may include helicopter type devices (e.g., quad copters), rolling helicopter type devices (e.g., roller copter devices that can fly and also roll along the ground, walls, or ceiling) and land vehicle type devices (e.g., automated cars that drive around a home). In some cases, the robotic devices **790** may be devices that are intended for other purposes and merely associated with the system **700** for use in appropriate circumstances. For instance, a robotic vacuum cleaner device may be associated with the monitoring system **700** as one of the robotic devices **790** and may be controlled to take action responsive to monitoring system events.

[0082] In some examples, the robotic devices **790** automatically navigate within a home. In these examples, the robotic devices **790** include sensors and control processors that guide movement of the robotic devices **790** within the home. For instance, the robotic devices **790** may navigate within the home using one or more cameras, one or more proximity sensors, one or more gyroscopes, one or more accelerometers, one or more magnetometers, a global positioning system (GPS) unit, an altimeter, one or more sonar or laser sensors, and/or any other types of sensors that aid in

navigation about a space. The robotic devices **790** may include control processors that process output from the various sensors and control the robotic devices **790** to move along a path that reaches the desired destination and avoids obstacles. In this regard, the control processors detect walls or other obstacles in the home and guide movement of the robotic devices **790** in a manner that avoids the walls and other obstacles.

[0083] In addition, the robotic devices **790** may store data that describes attributes of the home. For instance, the robotic devices **790** may store a floorplan and/or a three-dimensional model of the home that enables the robotic devices **790** to navigate the home. During initial configuration, the robotic devices **790** may receive the data describing attributes of the home, determine a frame of reference to the data (e.g., a home or reference location in the home), and navigate the home based on the frame of reference and the data describing attributes of the home. Further, initial configuration of the robotic devices **790** also may include learning of one or more navigation patterns in which a user provides input to control the robotic devices **790** to perform a specific navigation action (e.g., fly to an upstairs bedroom and spin around while capturing video and then return to a home charging base). In this regard, the robotic devices **790** may learn and store the navigation patterns such that the robotic devices **790** may automatically repeat the specific navigation actions upon a later request.

[0084] In some examples, the robotic devices **790** may include data capture and recording devices. In these examples, the robotic devices **790** may include one or more cameras, one or more motion sensors, one or more microphones, one or more biometric data collection tools, one or more temperature sensors, one or more humidity sensors, one or more air flow sensors, and/or any other types of sensors that may be useful in capturing monitoring data related to the home and users in the home. The one or more biometric data collection tools may be configured to collect biometric samples of a person in the home with or without contact of the person. For instance, the biometric data collection tools may include a fingerprint scanner, a hair sample collection tool, a skin cell collection tool, and/or any other tool that allows the robotic devices **790** to take and store a biometric sample that can be used to identify the person (e.g., a biometric sample with DNA that can be used for DNA testing).

[0085] In some implementations, the robotic devices **790** may include output devices. In these implementations, the robotic devices **790** may include one or more displays, one or more speakers, and/or any type of output devices that allow the robotic devices **790** to communicate information to a nearby user.

[0086] The robotic devices **790** also may include a communication module that enables the robotic devices **790** to communicate with the control unit **710**, each other, and/or other devices. The communication module may be a wireless communication module that allows the robotic devices **790** to communicate wirelessly. For instance, the communication module may be a Wi-Fi module that enables the robotic devices **790** to communicate over a local wireless network at the home. The communication module further may be a 900 MHZ wireless communication module that enables the robotic devices **790** to communicate directly with the control unit **710**. Other types of short-range wireless communication protocols, such as Bluetooth, Bluetooth LE, Z-wave, Zigbee, etc., may be used to allow the robotic devices **790** to communicate with other devices in the home. In some implementations, the robotic devices **790** may communicate with each other or with other devices of the system **700** through the network **705**.

[0087] The robotic devices **790** further may include processor and storage capabilities. The robotic devices **790** may include any suitable processing devices that enable the robotic devices **790** to operate applications and perform the actions described throughout this disclosure. In addition, the robotic devices **790** may include solid state electronic storage that enables the robotic devices **790** to store applications, configuration data, collected sensor data, and/or any other type of information available to the robotic devices **790**.

[0088] The robotic devices **790** are associated with one or more charging stations. The charging

stations may be located at predefined home base or reference locations in the home. The robotic devices **790** may be configured to navigate to the charging stations after completion of tasks needed to be performed for the monitoring system **700**. For instance, after completion of a monitoring operation or upon instruction by the control unit **710**, the robotic devices **790** may be configured to automatically fly to and land on one of the charging stations. In this regard, the robotic devices **790** may automatically maintain a fully charged battery in a state in which the robotic devices **790** are ready for use by the monitoring system **700**.

[0089] The charging stations may be contact based charging stations and/or wireless charging stations. For contact based charging stations, the robotic devices **790** may have readily accessible points of contact that the robotic devices **790** are capable of positioning and mating with a corresponding contact on the charging station. For instance, a helicopter type robotic device may have an electronic contact on a portion of its landing gear that rests on and mates with an electronic pad of a charging station when the helicopter type robotic device lands on the charging station. The electronic contact on the robotic device may include a cover that opens to expose the electronic contact when the robotic device is charging and closes to cover and insulate the electronic contact when the robotic device is in operation.

[0090] For wireless charging stations, the robotic devices **790** may charge through a wireless exchange of power. In these cases, the robotic devices **790** need only locate themselves closely enough to the wireless charging stations for the wireless exchange of power to occur. In this regard, the positioning needed to land at a predefined home base or reference location in the home may be less precise than with a contact based charging station. Based on the robotic devices **790** landing at a wireless charging station, the wireless charging station outputs a wireless signal that the robotic devices **790** receive and convert to a power signal that charges a battery maintained on the robotic devices **790**.

[0091] In some implementations, each of the robotic devices **790** has a corresponding and assigned charging station such that the number of robotic devices **790** equals the number of charging stations. In these implementations, the robotic devices **790** always navigate to the specific charging station assigned to that robotic device. For instance, a first robotic device may always use a first charging station and a second robotic device may always use a second charging station.

[0092] In some examples, the robotic devices **790** may share charging stations. For instance, the robotic devices **790** may use one or more community charging stations that are capable of charging multiple robotic devices **790**. The community charging station may be configured to charge multiple robotic devices **790** in parallel. The community charging station may be configured to charge multiple robotic devices **790** in serial such that the multiple robotic devices **790** take turns charging and, when fully charged, return to a predefined home base or reference location in the home that is not associated with a charger. The number of community charging stations may be less than the number of robotic devices **790**.

[0093] Also, the charging stations may not be assigned to specific robotic devices **790** and may be capable of charging any of the robotic devices **790**. In this regard, the robotic devices **790** may use any suitable, unoccupied charging station when not in use. For instance, when one of the robotic devices **790** has completed an operation or is in need of battery charge, the control unit **710** references a stored table of the occupancy status of each charging station and instructs the robotic device to navigate to the nearest charging station that is unoccupied.

[0094] The system **700** further includes one or more integrated security devices **780**. The one or more integrated security devices may include any type of device used to provide alerts based on received sensor data. For instance, the one or more control units **710** may provide one or more alerts to the one or more integrated security input/output devices **780**. Additionally, the one or more control units **710** may receive one or more sensor data from the sensors **720** and determine whether to provide an alert to the one or more integrated security input/output devices **780**.

[0095] The sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**,

and the integrated security devices **780** may communicate with the controller **712** over communication links **724, 726, 728, 732, 738, 736, and 784**. The communication links **724, 726, 728, 732, 738, and 784** may be a wired or wireless data pathway configured to transmit signals from the sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**, and the integrated security devices **780** to the controller **712**. The sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**, and the integrated security devices **780** may continuously transmit sensed values to the controller **712**, periodically transmit sensed values to the controller **712**, or transmit sensed values to the controller **712** in response to a change in a sensed value.

[0096] The communication links **724, 726, 728, 732, 738, and 784** may include a local network. The sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**, and the integrated security devices **780**, and the controller **712** may exchange data and commands over the local network. The local network may include 802.11 “Wi-Fi” wireless Ethernet (e.g., using low-power Wi-Fi chipsets), Z-Wave, Zigbee, Bluetooth, “Homeplug” or other “Powerline” networks that operate over AC wiring, and a Category 5 (CAT5) or Category 6 (CAT6) wired Ethernet network. The local network may be a mesh network constructed based on the devices connected to the mesh network.

[0097] The monitoring server **760** is an electronic device configured to provide monitoring services by exchanging electronic communications with the control unit **710**, the one or more user devices **740 and 750**, and the central alarm station server **770** over the network **705**. For example, the monitoring server **760** may be configured to monitor events (e.g., alarm events) generated by the control unit **710**. In this example, the monitoring server **760** may exchange electronic communications with the network module **714** included in the control unit **710** to receive information regarding events (e.g., alerts) detected by the control unit **710**. The monitoring server **760** also may receive information regarding events (e.g., alerts) from the one or more user devices **740 and 750**.

[0098] In some examples, the monitoring server **760** may route alert data received from the network module **714** or the one or more user devices **740 and 750** to the central alarm station server **770**. For example, the monitoring server **760** may transmit the alert data to the central alarm station server **770** over the network **705**.

[0099] The monitoring server **760** may store sensor and image data received from the monitoring system and perform analysis of sensor and image data received from the monitoring system. Based on the analysis, the monitoring server **760** may communicate with and control aspects of the control unit **710** or the one or more user devices **740 and 750**.

[0100] The monitoring server **760** may provide various monitoring services to the system **700**. For example, the monitoring server **760** may analyze the sensor, image, and other data to determine an activity pattern of a resident of the home monitored by the system **700**. In some implementations, the monitoring server **760** may analyze the data for alarm conditions or may determine and perform actions at the home by issuing commands to one or more of the controls **722**, possibly through the control unit **710**.

[0101] The central alarm station server **770** is an electronic device configured to provide alarm monitoring service by exchanging communications with the control unit **710**, the one or more mobile devices **740 and 750**, and the monitoring server **760** over the network **705**. For example, the central alarm station server **770** may be configured to monitor alerting events generated by the control unit **710**. In this example, the central alarm station server **770** may exchange communications with the network module **714** included in the control unit **710** to receive information regarding alerting events detected by the control unit **710**. The central alarm station server **770** also may receive information regarding alerting events from the one or more mobile devices **740 and 750** and/or the monitoring server **760**.

[0102] The central alarm station server **770** is connected to multiple terminals **772 and 774**. The

terminals **772** and **774** may be used by operators to process alerting events. For example, the central alarm station server **770** may route alerting data to the terminals **772** and **774** to enable an operator to process the alerting data. The terminals **772** and **774** may include general-purpose computers (e.g., desktop personal computers, workstations, or laptop computers) that are configured to receive alerting data from a server in the central alarm station server **770** and render a display of information based on the alerting data. For instance, the controller **712** may control the network module **714** to transmit, to the central alarm station server **770**, alerting data indicating that a sensor **720** detected motion from a motion sensor via the sensors **720**. The central alarm station server **770** may receive the alerting data and route the alerting data to the terminal **772** for processing by an operator associated with the terminal **772**. The terminal **772** may render a display to the operator that includes information associated with the alerting event (e.g., the lock sensor data, the motion sensor data, the contact sensor data, etc.) and the operator may handle the alerting event based on the displayed information.

[0103] In some implementations, the terminals **772** and **774** may be mobile devices or devices designed for a specific function. Although FIG. 7 illustrates two terminals for brevity, actual implementations may include more (and, perhaps, many more) terminals.

[0104] The one or more authorized user devices **740** and **750** are devices that host and display user interfaces. For instance, the user device **740** is a mobile device that hosts or runs one or more native applications (e.g., the smart home application **742**). The user device **740** may be a cellular phone or a non-cellular locally networked device with a display. The user device **740** may include a cell phone, a smart phone, a tablet PC, a personal digital assistant (“PDA”), or any other portable device configured to communicate over a network and display information. For example, implementations may also include Blackberry-type devices (e.g., as provided by Research in Motion), electronic organizers, iPhone-type devices (e.g., as provided by Apple), iPod devices (e.g., as provided by Apple) or other portable music players, other communication devices, and handheld or portable electronic devices for gaming, communications, and/or data organization. The user device **740** may perform functions unrelated to the monitoring system, such as placing personal telephone calls, playing music, playing video, displaying pictures, browsing the Internet, maintaining an electronic calendar, etc.

[0105] The user device **740** includes a smart home application **742**. The smart home application **742** refers to a software/firmware program running on the corresponding mobile device that enables the user interface and features described throughout. The user device **740** may load or install the smart home application **742** based on data received over a network or data received from local media. The smart home application **742** runs on mobile devices platforms, such as iPhone, iPod touch, Blackberry, Google Android, Windows Mobile, etc. The smart home application **742** enables the user device **740** to receive and process image and sensor data from the monitoring system.

[0106] The user device **750** may be a general-purpose computer (e.g., a desktop personal computer, a workstation, or a laptop computer) that is configured to communicate with the monitoring server **760** and/or the control unit **710** over the network **705**. The user device **750** may be configured to display a smart home user interface **752** that is generated by the user device **750** or generated by the monitoring server **760**. For example, the user device **750** may be configured to display a user interface (e.g., a web page) provided by the monitoring server **760** that enables a user to perceive images captured by the camera **730** and/or reports related to the monitoring system. Although FIG. 5 illustrates two user devices for brevity, actual implementations may include more (and, perhaps, many more) or fewer user devices.

[0107] In some implementations, the one or more user devices **740** and **750** communicate with and receive monitoring system data from the control unit **710** using the communication link **738**. For instance, the one or more user devices **740** and **750** may communicate with the control unit **710** using various local wireless protocols such as Wi-Fi, Bluetooth, Z-wave, Zigbee, HomePlug

(ethernet over power line), or wired protocols such as Ethernet and USB, to connect the one or more user devices **740** and **750** to local security and automation equipment. The one or more user devices **740** and **750** may connect locally to the monitoring system and its sensors and other devices. The local connection may improve the speed of status and control communications because communicating through the network **705** with a remote server (e.g., the monitoring server **760**) may be significantly slower.

[0108] Although the one or more user devices **740** and **750** are shown as communicating with the control unit **710**, the one or more user devices **740** and **750** may communicate directly with the sensors and other devices controlled by the control unit **710**. In some implementations, the one or more user devices **740** and **750** replace the control unit **710** and perform the functions of the control unit **710** for local monitoring and long range/offsite communication.

[0109] In other implementations, the one or more user devices **740** and **750** receive monitoring system data captured by the control unit **710** through the network **705**. The one or more user devices **740**, **750** may receive the data from the control unit **710** through the network **705** or the monitoring server **760** may relay data received from the control unit **710** to the one or more user devices **740** and **750** through the network **705**. In this regard, the monitoring server **760** may facilitate communication between the one or more user devices **740** and **750** and the monitoring system.

[0110] In some implementations, the one or more user devices **740** and **750** may be configured to switch whether the one or more user devices **740** and **750** communicate with the control unit **710** directly (e.g., through link **738**) or through the monitoring server **760** (e.g., through network **705**) based on a location of the one or more user devices **740** and **750**. For instance, when the one or more user devices **740** and **750** are located close to the control unit **710** and in range to communicate directly with the control unit **710**, the one or more user devices **740** and **750** use direct communication. When the one or more user devices **740** and **750** are located far from the control unit **710** and not in range to communicate directly with the control unit **710**, the one or more user devices **740** and **750** use communication through the monitoring server **760**.

[0111] Although the one or more user devices **740** and **750** are shown as being connected to the network **705**, in some implementations, the one or more user devices **740** and **750** are not connected to the network **705**. In these implementations, the one or more user devices **740** and **750** communicate directly with one or more of the monitoring system components and no network (e.g., Internet) connection or reliance on remote servers is needed.

[0112] In some implementations, the one or more user devices **740** and **750** are used in conjunction with only local sensors and/or local devices in a house. In these implementations, the system **700** includes the one or more user devices **740** and **750**, the sensors **720**, the home automation controls **722**, the camera **730**, the robotic devices **790**, and the UAV control engine **757**. The one or more user devices **740** and **750** receive data directly from the sensors **720**, the home automation controls **722**, the camera **730**, the robotic devices **790**, and the UAV control engine **757** and sends data directly to the sensors **720**, the home automation controls **722**, the camera **730**, the robotic devices **790**, and the UAV control engine **757**. The one or more user devices **740**, **750** provide the appropriate interfaces/processing to provide visual surveillance and reporting.

[0113] In other implementations, the system **700** further includes network **705** and the sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**, the robotic devices **790**, and the UAV control engine **757** are configured to communicate sensor and image data to the one or more user devices **740** and **750** over network **705** (e.g., the Internet, cellular network, etc.). In yet another implementation, the sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**, the robotic devices **790**, and the UAV control engine **757** (or a component, such as a bridge/router) are intelligent enough to change the communication pathway from a direct local pathway when the one or more user devices **740** and **750** are in close physical proximity to the sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**, the robotic

devices **790**, and the UAV control engine **757** to a pathway over network **705** when the one or more user devices **740** and **750** are farther from the sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**, the robotic devices **790**, and the control engine.

[0114] In some examples, the system leverages GPS information from the one or more user devices **740** and **750** to determine whether the one or more user devices **740** and **750** are close enough to the sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**, the robotic devices **790**, and the UAV control engine **757** to use the direct local pathway or whether the one or more user devices **740** and **750** are far enough from the sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**, the robotic devices **790**, and the UAV control engine **757** that the pathway over network **705** is required.

[0115] In other examples, the system leverages status communications (e.g., pinging) between the one or more user devices **740** and **750** and the sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**, the robotic devices **790**, and the UAV control engine **757** to determine whether communication using the direct local pathway is possible. If communication using the direct local pathway is possible, the one or more user devices **740** and **750** communicate with the sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**, the robotic devices **790**, and the UAV control engine **757** using the direct local pathway. If communication using the direct local pathway is not possible, the one or more user devices **740** and **750** communicate with the sensors **720**, the home automation controls **722**, the camera **730**, the thermostat **734**, the robotic devices **790**, and the UAV control engine **757** using the pathway over network **705**.

[0116] In some implementations, the system **700** provides end users with access to images captured by the camera **730** to aid in decision making. The system **700** may transmit the images captured by the camera **730** over a wireless WAN network to the user devices **740** and **750**. Because transmission over a wireless WAN network may be relatively expensive, the system **700** can use several techniques to reduce costs while providing access to significant levels of useful visual information (e.g., compressing data, down-sampling data, sending data only over inexpensive LAN connections, or other techniques).

[0117] In some implementations, a state of the monitoring system and other events sensed by the monitoring system may be used to enable/disable video/image recording devices (e.g., the camera **730**). In these implementations, the camera **730** may be set to capture images on a periodic basis when the alarm system is armed in an “away” state, but set not to capture images when the alarm system is armed in a “home” state or disarmed. In addition, the camera **730** may be triggered to begin capturing images when the alarm system detects an event, such as an alarm event, a door-opening event for a door that leads to an area within a field of view of the camera **730**, or motion in the area within the field of view of the camera **730**. In other implementations, the camera **730** may capture images continuously, but the captured images may be stored or transmitted over a network when needed.

[0118] The described systems, methods, and techniques may be implemented in digital electronic circuitry, computer hardware, firmware, software, or in combinations of these elements. Apparatus implementing these techniques may include appropriate input and output devices, a computer processor, and a computer program product tangibly embodied in a machine-readable storage device for execution by a programmable processor. A process implementing these techniques may be performed by a programmable processor executing a program of instructions to perform desired functions by operating on input data and generating appropriate output. The techniques may be implemented in one or more computer programs that are executable on a programmable system including at least one programmable processor coupled to receive data and instructions from, and to transmit data and instructions to, a data storage system, at least one input device, and at least one output device.

[0119] Each computer program may be implemented in a high-level procedural or object-oriented

programming language, or in assembly or machine language if desired; and in any case, the language may be a compiled or interpreted language. Suitable processors include, by way of example, both general and special purpose microprocessors. Generally, a processor will receive instructions and data from a read-only memory and/or a random access memory.

[0120] Storage devices suitable for tangibly embodying computer program instructions and data include all forms of non-volatile memory, including by way of example semiconductor memory devices, such as Erasable Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM), and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and Compact Disc Read-Only Memory (CD-ROM). Any of the foregoing may be supplemented by, or incorporated in, specially designed ASICs (application-specific integrated circuits).

[0121] It will be understood that various modifications may be made. For example, other useful implementations could be achieved if steps of the disclosed techniques were performed in a different order and/or if components in the disclosed systems were combined in a different manner and/or replaced or supplemented by other components. Accordingly, other implementations are within the scope of the disclosure.

Claims

1. (canceled)

2. A system comprising one or more computers and one or more storage devices on which are stored instructions that are operable, when executed by the one or more computers, to cause the one or more computers to perform operations comprising: detecting, using one or more images captured by a camera, a boundary between a first object and a second object both of which are located in a physical area; in response to detecting the boundary between the first object and the second object both of which are located in a physical area, computing, for the camera, one or more fields of view for the camera and of the physical area; predicting, for at least one of the one or more fields of view for the camera, a portion of the physical area that would be depicted in an image captured by the camera when the camera has the corresponding field of view; computing an adjustment for the camera using the predicted portion of the physical area that would be depicted in an image captured by the camera when the camera has the corresponding field of view; and sending one or more instructions to cause a change in a position of the camera using the adjustment.

3. The system of claim 2, wherein: detecting the boundary comprises detecting one or more attributes of the boundary between the first object and the second object; and computing the adjustment uses the one or more attributes of the boundary between the first object and the second object.

4. The system of claim 2, wherein sending the one or more instructions causes a change in a position or orientation of the camera.

5. The system of claim 3, wherein sending the one or more instructions causes a change in a position or orientation of a drone to which the camera is coupled.

6. The system of claim 2, wherein detecting the boundary between the first object and the second object comprises detecting a physical boundary between a first property and a second property.

7. The system of claim 2, the operations comprising: determining to reduce an amount an object is depicted in a current field of view of the camera, the object comprising the first object or the second object, wherein: computing the one or more fields of view for the camera and of the physical area is responsive to determining to reduce the amount the object is depicted in the current field of view of the camera.

8. The system of claim 2, wherein: the system comprises the camera; and the one or more computers comprise on board circuitry in the system.

9. The system of claim 2, wherein the system comprises a drone.

- 10.** The system of claim 2, wherein the first object comprises a first property and the second object comprises a second, neighboring property.
- 11.** A computer-implemented method comprising: detecting, using one or more images captured by a camera, a boundary between a first object and a second object both of which are located in a physical area; in response to detecting the boundary between the first object and the second object both of which are located in a physical area, computing, for the camera, one or more fields of view for the camera and of the physical area; predicting, for at least one of the one or more fields of view for the camera, a portion of the physical area that would be depicted in an image captured by the camera when the camera has the corresponding field of view; computing an adjustment for the camera using the predicted portion of the physical area that would be depicted in an image captured by the camera when the camera has the corresponding field of view; and sending one or more instructions to cause a change in a position of the camera using the adjustment.
- 12.** The method of claim 11, wherein: detecting the boundary comprises detecting one or more attributes of the boundary between the first object and the second object; and computing the adjustment uses the one or more attributes of the boundary between the first object and the second object.
- 13.** The method of claim 11, wherein sending the one or more instructions causes a change in a position or orientation of the camera.
- 14.** The method of claim 13, wherein sending the one or more instructions causes a change in a position or orientation of a drone to which the camera is coupled.
- 15.** The method of claim 11, wherein detecting the boundary between the first object and the second object comprises detecting a physical boundary between a first property and a second property.
- 16.** The method of claim 11, comprising: determining to reduce an amount an object is depicted in a current field of view of the camera, the object comprising the first object or the second object, wherein: computing the one or more fields of view for the camera and of the physical area is responsive to determining to reduce the amount the object is depicted in the current field of view of the camera.
- 17.** The method of claim 11, wherein: the system comprises the camera; and the one or more computers comprise on board circuitry in the system.
- 18.** The method of claim 11, wherein the system comprises a drone.
- 19.** The method of claim 11, wherein the first object comprises a first property and the second object comprises a second, neighboring property.
- 20.** One or more non-transitory computer storage media encoded with instructions that, when executed by one or more computers, cause the one or more computers to perform operations comprising: detecting, using one or more images captured by a camera, a boundary between a first object and a second object both of which are located in a physical area; in response to detecting the boundary between the first object and the second object both of which are located in a physical area, computing, for the camera, one or more fields of view for the camera and of the physical area; predicting, for at least one of the one or more fields of view for the camera, a portion of the physical area that would be depicted in an image captured by the camera when the camera has the corresponding field of view; computing an adjustment for the camera using the predicted portion of the physical area that would be depicted in an image captured by the camera when the camera has the corresponding field of view; and sending one or more instructions to cause a change in a position of the camera using the adjustment.
- 21.** The media of claim 20, wherein: detecting the boundary comprises detecting one or more attributes of the boundary between the first object and the second object; and computing the adjustment uses the one or more attributes of the boundary between the first object and the second object.
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